2.5.4 Conclusion

The current work presents two improved RMG mechanisms for describing the flame inhibition chemistry of 2-BTP in methane flames, one obtained through manual modifications (MM) and the other through database updates incorporating higher-accuracy halocarbon-related chemical data (UD). In the case of the improved RMG (MM) model, analysis of the initial iteration of this improved mechanism prompted three model modifications: the inclusion of pressure dependent chemistry, incorporation of the BrOH species, and altering the kinetics of reactions $Br + CH_3 \rightleftharpoons CH_3Br$ and $CHF_3 \rightleftharpoons HF + CF_2$. In the case of the improved RMG (UD) model, thermochemical and kinetic updates to the database eliminated the need for any manual post-generation modifications.

The improved models were used to predict laminar burning velocities for methane/normal air systems with variable initial volume fractions of 2-BTP. Compared to the previously presented RMG model [102], the improved models demonstrate enhanced quantitative agreement with published experimental data and laminar burning velocities predicted using the 2015 NIST 2-BTP mechanism, with the improved RMG (UD) model demonstrating the closest overall agreement with experiments. Both the improved RMG models are significantly smaller in size compared to the previous RMG mechanism.

This work utilizes the modified database that I have developed, one that integrates more halocarbon-specific chemistry into the existing RMG framework. This work also calls attention to a technical issue that users may experience when generating a mechanism in the RMG software over prolonged periods of time. To resolve this, an idealized method for model generation is described.

2.6 Halocarbon Blending Methods

2.6.1 Introduction

While the RMG database had initially only included C, H, O, N, and S chemistry, its scope has recently been expanded to include halogen (fluorine, chlorine, bromine) combustion chemistry [102], as well as difluoromethane combustion chemistry (Section 2.3) and the degradation chemistry of per- and polyfluoroalkyl substances (PFAS) (Section 3.1), collectively enabling the automatic generation of higher-accuracy halocarbon combustion models. These capabilities have already been validated across several halocarbon systems: RMG has previously been used to generate individual kinetic mechanisms for the combustion of seven pure halocarbon refrigerants (C_2H_5F, CH_2F_2, CH_2FCH_2F, CH_2FCHF_2, CH_2CF_3, CH_3CHF_2, and CH_3F) in air [13] (Figure 2.1), demonstrating

good agreement with experimental values [13], and to construct a flame suppression model for 2-BTP ($CH_2=CBrCF_3$) in methane flames [102] (Section 2.5). However, these efforts were primarily proof-of-concept demonstrations of RMG's ability to model halocarbon species; none examined refrigerant blends or how different blending strategies within RMG influenced the combustion behavior included in the final mechanism—a gap this work aims to address.

Within RMG, multi-component blends can be generated in several different ways—some intuitive, others considerably less so—each discussed in turn below. The first approach involves simple concatenation: an n-component blend mechanism can be constructed by using RMG to generate individual combustion mechanisms for each of the n pure halocarbons, then concatenating these n models into a single mechanism encompassing all components. Another approach involves generating the blend mechanism in a single RMG run, in which all refrigerant components of the blend are specified as starting reactants in the input file. Lastly, a blend mechanism in RMG can be constructed by first generating separate combustion mechanisms for each pure component, similar to the initial step of the first approach. Rather than concatenating these pure models afterward, the species from each can instead be seeded as initial core species in a new RMG job used to generate the multi-component blend mechanism. Prior studies constructing blend mechanisms in RMG have typically relied on concatenation [148, 149], the inclusion of all blend components as initial reactants [150, 151], or a hybrid of the two approaches [152]. The final approach—building pure models first, then seeding the blend model with all species from those pure models—is rarely used, as it tends to be the most time-consuming and least intuitive of the three.

This work investigates three distinct approaches to binary blend construction in RMG: (1) seeding the initial RMG core with species from both pure-component models and generating the blend mechanism de novo; (2) limiting the seeding of the initial RMG core to the two blend components and the oxidizer, and (3) appending the kinetic models of the two pure components of the binary blend together. The blend mechanisms generated using each method were then used to predict laminar burning velocities across systems with increasing refrigerant volume fractions in air, and the resulting predictions were compared. Variations in predicted laminar burning velocities across mechanisms generated by different blending methods are expected to arise from disparities in the incorporated chemistry, as some methods are likely better suited to capturing blend-specific cross-reactions and interaction chemistry—the key chemical signature distinguishing a true blend mechanism from a simple superposition of pure-component mechanisms. The overall goal of this project is to investigate whether the choice of blending method affects flame speed predictions for the refrigerant blend system, specifically examining how different blending methods in RMG influence

the quality of the resulting blend model. It is important to clarify that the primary objective of this study is not to identify ideal refrigerant blend alternatives, but rather to demonstrate and refine a computational workflow using RMG that can ultimately be applied to achieve this goal.

2.6.2 Methodology

2.6.2.1 Model Generation in RMG

This study investigates three RMG-based blending methods: (1) simple concatenation ("SC"), in which individual combustion mechanisms are generated separately for each pure blend component and subsequently merged into a single mechanism representing the blend; (2) RMG-minimal seeded ("RMG-MS"), in which the initial RMG core is limited to only the blend's two components and the oxidizer; and (3) RMG-pure model seeded ("RMG-PMS"), in which the blend mechanism is generated using a core seeded with species drawn from the individual pure-component models. Figure 2.40 provides a conceptual illustration of these blending techniques.

Binary refrigerant blends of hydrocarbons (R50/methane, R290/propane, R600a/isobutane) and HFCs (R32/CH_2F_2, R134a/CH_2F-CF_3, R152a/CH_3-CHF_2) were generated in RMG, yielding nine blend combinations in total. For select blend systems, experimentally measured flame speeds were available in the literature and used for direct comparison against simulated predictions.

The publicly available `PFAS_JPCA_paper` branches of the official RMG-database and RMG-Py repositories (Git commits 55117dda and c30f89a5, respectively) were used to generate all mechanisms in this work. These commits represent the final versions of the software developed over the course of this dissertation, incorporating higher-accuracy kinetics and thermochemistry for halocarbon and PFAS chemistry; accordingly, they were adopted for use in this work. RMG input files defined the following conditions and specifications: a temperature range of 800–2000 K, a pressure range of 1–10 bar, the incorporation of pressure-dependent chemistry, and the inclusion of hydrocarbon and halocarbon-specific thermochemical and kinetic libraries (presented in references [102] and developed in the projects of this dissertation).

For pure refrigerant mechanisms, RMG input files specified the initial reactants as the single refrigerant component along with oxygen and nitrogen. The specified initial reactant ratios spanned $\pm 20\%$ around the stoichiometric composition for combustion of the single refrigerant in air. For hydrocarbon refrigerants, complete stoichiometric combustion yields CO_2 and H_2O as products [153] (e.g. R290/propane: $C_3H_8 + 5\,(O_2 + 3.76\,N_2) \longrightarrow 3\,CO_2 + 4\,H_2O + 18.8\,N_2$). Stoichiometric combustion of halocarbon refrigerants containing at least as many hydrogen atoms

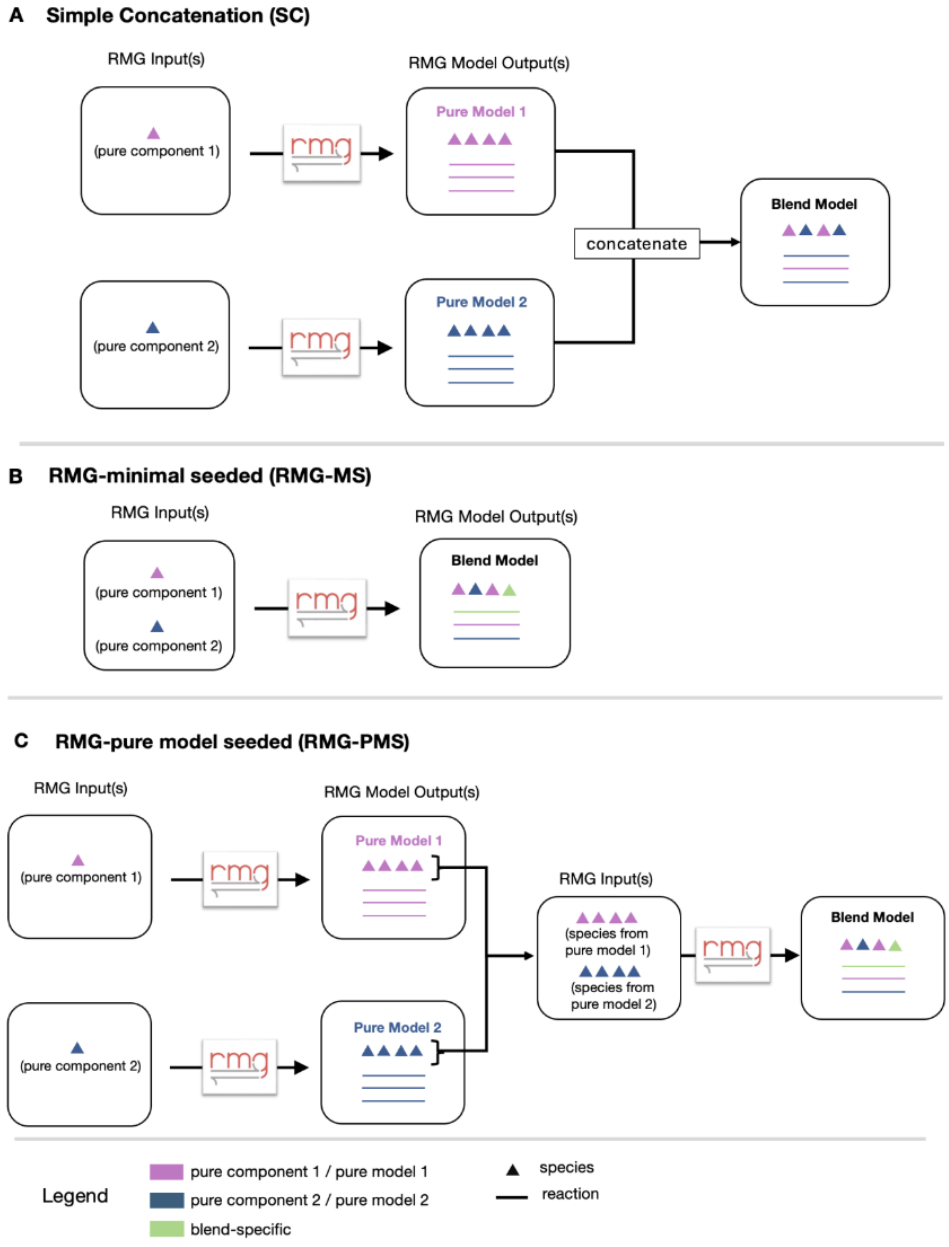


Figure 2.40: Schematic overview of the three RMG blend generation methods investigated in the present work. **(A)** Simple Concatenation (SC): individual mechanisms are constructed for each pure component of the binary blend, then concatenated to produce the final blend mechanism, **(B)** RMG-minimal seeded (RMG-MS): the blend mechanism is generated using an RMG core seeded with just the two blend component species (and the oxidizer, not shown), and **(C)** RMG-pure model seeded (RMG-PMS): the blend mechanism is generated using an RMG core pre-populated with all species from both pure-component models.

as fluorine atoms was predicted to produce HF to capture the fluorine content, with any remaining hydrogen forming H_2O and all carbon converted to CO_2 [153] (e.g. R32/difluoromethane: $CH_2F_2 + (O_2 + 3.76\,N_2) \longrightarrow CO_2 + 2\,HF + 3.76\,N_2$). In cases where the number of fluorine atoms exceeded the number of hydrogen atoms in the refrigerant species, carbonyl fluoride (COF_2) was predicted as an additional combustion product, capturing the fluorine atoms not accounted for by HF formation [153] (e.g. R1234yf: $C_3H_2F_4 + 2.5\,(O_2 + 3.76\,N_2) \longrightarrow 2\,CO_2 + 2\,HF + COF_2 + 9.4\,N_2$).

In addition to pure-component systems, 50/50 binary blend mixtures were generated in RMG, with each blending technique requiring a distinct RMG input configuration. For blend mechanisms generated via the SC method, the two pure-component mechanisms were concatenated using the `MergeModels` tool available within the RMG software, which combines two RMG-generated models into a single merged mechanism file, identifying equivalent species and eliminating redundant reactions between the two mechanisms to be combined. In this process, the first model generally takes priority, with thermochemistry and kinetics for any species or reactions common to both models drawn exclusively from that first model rather than the second. This distinction is inconsequential for the present work, however, as both mechanisms being merged were constructed using the same RMG database and therefore contain identical thermochemical and kinetic parameters for any shared species and reactions. While the resulting concatenated blend mechanism encompasses all species and reactions present in the two pure models, it lacks any blend-specific chemistry that could arise from interactions between species unique to each of the two pure mechanisms.

For blend mechanisms generated via the RMG-MS method, the RMG input file specifies only four species in the initial RMG core (the two components of the binary blend, along with oxygen and nitrogen) and the reactants' initial molar composition ranges from 20% below to 20% above the stoichiometric composition for combustion of a 50/50 blend in normal air. Lastly, for input files used to generate blend mechanisms via the RMG-PMS method, all core species from the two pure-component mechanisms comprising the blend were identified and explicitly specified in the blend input file. The initial molar compositions were defined identically to the RMG-MS case; notably, although the RMG-PMS method includes all species from both pure models in the initial RMG core, the actual reactants present at the start of the simulation remain limited to the two blend components, along with oxygen and nitrogen. It is anticipated that blend models generated using the RMG-PMS and -MS methods will capture blend-specific chemistry, such as cross reactions, as this approach allows RMG to explore reaction pathways between species from both blend components during mechanism generation, rather than treating each component's chemistry in isolation.

Mechanism generation time for these complex reactive systems is often extensive, some-

times taking weeks to meet the user-defined completion criteria in RMG. However, the majority of essential chemistry is typically identified early in the RMG run, with less critical reactions gradually added as the simulation progresses over several days—a presumption that can be confirmed by monitoring the convergence of predicted LBV over runtime, as was done in this study. Each RMG simulation in the present work, whether simulating a pure or binary refrigerant system, was thus constrained to a maximum wall time of 10 days. This limit was chosen to reflect the typical computing and timeline constraints faced by a standard RMG user. Thus, blending methods were evaluated not only in terms of the comprehensiveness of the resulting mechanism, but also with respect to which approach is best suited to the computational resources available to the user.

2.6.2.2 Laminar Burning Velocity Prediction in Cantera

Laminar burning velocities (LBVs) for generated halocarbon mechanisms were computed using a freely-propagating, adiabatic premixed flame simulation in Cantera software. Pure and blended refrigerant systems were simulated at an initial temperature of 298 K and a pressure of 1 atm. LBVs for mechanisms were calculated across increasing total refrigerant fractions, up to 0.25, to observe blending effects. In addition, flame speed conditions corresponding to the experimental measurements published in 2024 by Zhang et al. (and previously discussed in Section 2.4.3) were simulated in Cantera using mechanisms constructed via each blending method, across a range of increasing equivalence ratios.

2.6.3 Results and discussion

To determine whether the chosen maximum wall time (10 days) was sufficient to produce a converged model, flame speed was monitored as a function of mechanism growth. Figure 2.41 illustrates the monitoring of flame speed convergence for the minimally-seeded (MS) binary blend models; at the chosen wall time limit (indicated by the vertical dotted line), simulated flame speed had stabilized for most MS blend models. Since the MS method represents the most computationally demanding of the three blending approaches—building each pure-component's submechanisms along with any relevant cross reactions from a minimal starting core rather than a pre-seeded or pre-concatenated set of species—its convergence within the 10-day limit suggests that this run time was likely sufficient to produce converged models across all three blending methods.

Figure 2.42 presents computed laminar burning velocities for both the pure-component mechanisms and the HC/HFC 50/50 binary blend mechanisms constructed via the concatenated

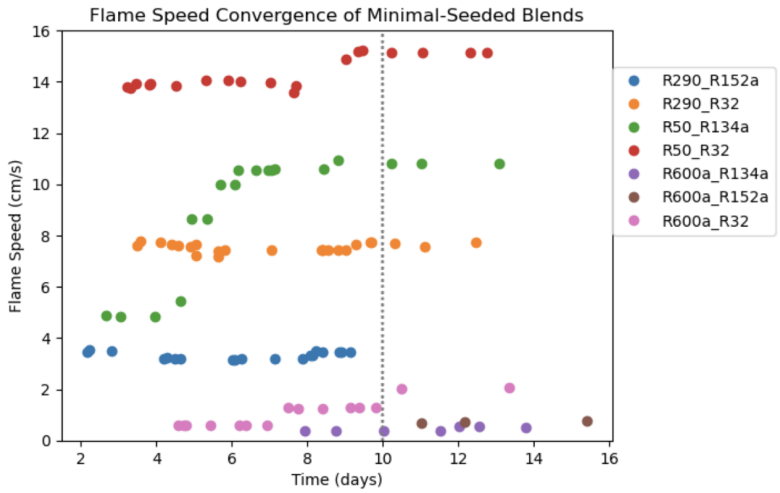


Figure 2.41: Simulated flame speed for minimally-seeded (MS) binary blend models as a function of run time, evaluated at an initial temperature of 298 K, 1 atm, and a total refrigerant volume fraction of 0.10. Convergence of the flame speed is taken as an indicator of model completeness. The vertical dotted line indicates the chosen run time limit.

and minimal seeded (MS) blending methods. Table 2.4 reports the number of species and reactions comprising each RMG mechanism. As expected, the pure HC systems (R290, R50, R600a) achieve faster maximum flame speeds than the pure HFC systems (R32, R134a, R152a), consistent with their respective ASHRAE 34 flammability classifications—R290, R50, and R600a are all rated Class 3 (highly flammable), whereas R32 and R152a are rated A2L (mildly flammable) and R134a is rated nonflammable (Class 1). Generally, the HC/HFC blend mechanisms achieve maximum flame speeds that fall between those of the corresponding pure-component models, although the concatenated and MS models often disagree on the magnitude of this maximum. In most of the tested cases, the concatenated and MS blend models predict distinct flame speeds, with the magnitude of this discrepancy varying by system. Differences in maximum flame speed among the blend models are greatest for the R290/R152a, R600a/R134a, R600a/R152a and R600a/R32 systems, with discrepancies of 13.9, 12.9, 14.8, and 21.4 cm/s, respectively. The remaining tested blends show substantially smaller discrepancies in maximum flame speed, with several falling below 1 cm/s—R290/R32, R50/R152a, and R50/R32, with differences of 0.11, 0.45, and 0.63 cm/s, respectively—and others falling below approximately 4 cm/s: R290/R134a and R50/R134a, with

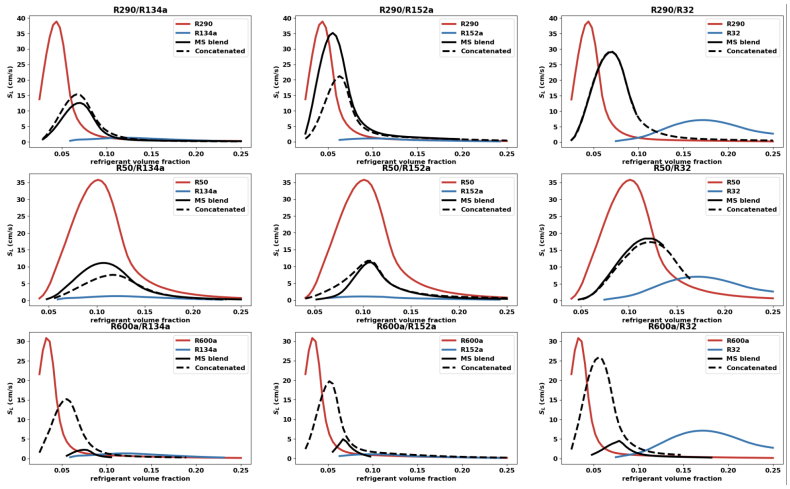


Figure 2.42: Laminar burning velocities of pure and 50/50 binary refrigerant mixtures by volume fraction combusted in normal air, with the x-axis representing total refrigerant volume fraction. Plots include predicted flame speeds from pure halocarbon mechanisms (solid red/blue), concatenated blend mechanisms (dotted black), and MS blended mechanisms (solid black).

differences of 2.7 and 3.5 cm/s respectively. Although nearly all blend cases exhibit differences in the magnitude of their maximum flame speeds, the overall flame speed profiles remain largely consistent across most blend cases, with the maximum flame speed occurring at the similar total refrigerant volume fractions for both the concatenated and MS blend mechanisms. The R600a/R134a, R600a/R152a and R600a/R32 blends are three notable exceptions, exhibiting markedly different flame speed profiles, including substantial differences in the ϕs at which the maximum flame speed is expected.

The large discrepancy in predicted flame speeds between each blend mechanism for the R600a-involving systems is unexpected; to verify these results, the MS blend model for the R600a/R32 system was regenerated using an identical process—the same 10-day run time limit, input file, RMG version, and high-performance computing resources. Figure 2.44 reports predicted flamespeeds from this newly regenerated R600a/R32 MS blend (labeled "MS (regenerated)" in the figure, represented by the dotted black line). Despite identical RMG inputs and run time limits, the regenerated MS blend flame speed profile bears a much closer resemblance to that of the concatenated model, and differs drastically from the originally generated MS blend. It was

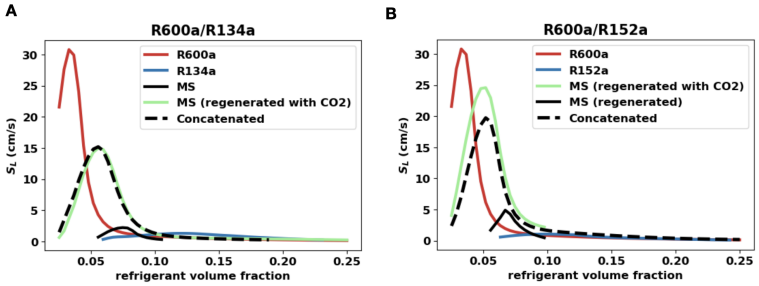


Figure 2.43: Laminar burning velocities of pure and blended refrigerant mixtures by volume fraction combusted in normal air, with the x-axis representing total refrigerant volume fraction. Plots include predicted flame speeds from pure halocarbon mechanisms (solid red/blue), concatenated blend mechanisms (dotted black), MS blended mechanisms (solid black) and regenerated MS blend with CO_2 explicitly defined in the RMG input file (solid light green). Speeds predicted for the **A)** R600a/R134a system and **B)** R600a/R152a system are both shown.

found that the regenerated MS blend included one additional species absent from the original MS blend—a species critical to the combustion system: CO_2. The original MS blend mechanism for the R600a/R32 system ended with 97 core species by the end of the 10-day run time limit; inspection of the RMG log files revealed that the 98th species—added just beyond this artificially imposed run time limit—was indeed CO_2. Without CO_2, the originally generated MS mechanism cannot capture the highly exothermic CO + OH $\rightleftharpoons$ H + CO_2 reaction—one of the primary heat-release and chain-propagation steps in combustion—resulting in a lower predicted flame temperature and, consequently, a lower flame speed. To confirm that CO_2 alone was responsible for the discrepancy, flame speeds were predicted using the version of the original MS mechanism that had grown to 98 core species just beyond the imposed 10-day limit; as shown in Figure 2.44, this 98-species model—now including CO_2—matches the flame speed profile of the regenerated MS model. In the other R600a-involving blend cases (R600a/R134a and R600a/R152a), the MS mechanisms similarly lack CO_2, resulting in significantly lower predicted flame speeds than their concatenated counterpart. All R600a-involving MS blend cases were regenerated with CO_2 explicitly defined in the input file to see if new MS blend profiles would similarly shift (like in the R600a/R32 case) to resemble the concatenated mechanism. Figure 2.43 reports predicted flame speeds for these two newly generated MS blends with CO_2, now initialized in the starting RMG core; flame speed profiles for both blend cases now map similarly with their concatenated analogues.

This finding raises an important question: how can an RMG model run for 10 days on high-performance computing resources without ever incorporating CO_2 into its mechanism? Diagnostic investigation traced this failure to a hardware issue rather than a limitation of RMG itself: the specific compute node on which these R600a-involving blend runs were executed was throttled, running roughly 75–100$\times$ slower than a healthy node of the same partition. For context, R600a-containing systems typically require approximately 150–200 RMG iterations before CO_2 is incorporated into the core; this number of iterations is usually reached within a few hours on a healthy node, but require 8–14 days on the throttled node. As a result, the imposed run time limit is exceeded before CO_2 could ever be added. In addition to the R600a-involving blends, two pure component models (R134a and R152a) additionally utilized this slow compute node, resulting in immature models by the end of the 10-day run time. This is clearly reflected in Table 2.4: every system that ran on the throttled node (R134a, R152a, R600a/R134a, R600a/R152a, and R600a/R32) incorporates substantially fewer species into its final model than the pure or blend systems run on non-throttled nodes. This result underscores that model comprehensiveness under a fixed wall-time budget is not solely a function of the blending method or RMG's underlying algorithm, but can also be strongly influenced by underlying computational infrastructure; accordingly, all affected runs were subsequently repeated on unaffected nodes to ensure a fair comparison between blending methods. Among the results reported in Figure 2.42, only two systems—R290/R32 and R50/R32—possess corresponding pure and MS or concatenated blend mechanisms that were unaffected by the throttled-node issue, and are therefore the only systems suitable for a fair comparison between the MS and concatenated blending methods.

Following the 10-day run time limit used to generate the pure-component mechanisms for each blend system, PMS blend mechanisms—which use species from the pure-component models comprising the blend to inform model generation—were subsequently run. However, these PMS blend mechanisms were similarly affected by the throttled node, resulting in an effective run time far shorter than the intended 10 days. Upon discovery of this issue, the affected PMS blends were resubmitted to unaffected compute nodes and are currently being regenerated.

Of this initial set of blend cases, only R290/R32 and R50/R32 were unaffected by the slow-node issue, with pure and MS/concatenated blended mechanisms having genuinely identical run times. In these two cases, the concatenated and MS blend mechanisms yield very similar flame speed predictions, despite substantial differences in mechanism size: the concatenated model includes 258 species for R290/R32 and 261 for R50/R32, compared to only 163 and 143 species, respectively, for the corresponding MS blends. This disparity reflects an inherent trade-off between the two methods. The concatenated method benefits from an effective run time of 20 days—the sum of the

two independent 10-day pure-component runs—allowing it to incorporate a substantially larger set of species and reactions into the final model; however, because it simply merges two independently generated pure-component mechanisms, it cannot discover new reactions arising specifically from interactions between species unique to each pure system. The MS method, by contrast, operates within a single 10-day run time budget, yielding a comparatively smaller mechanism, but its unified core/edge exploration allows it to identify genuine cross-reactions between species from both blend components that concatenation alone is structurally incapable of capturing. The similarity in flame speed predictions between the two methods in these two cases suggests that, for these stoichiometrically simple pure-hydrocarbon/HFC blends, any blend-specific cross-reactions captured by the MS method do not substantially alter the predicted flame speed relative to the larger (but cross-reaction-blind) concatenated mechanism. However, this conclusion cannot be extrapolated to any of the remaining blend systems, as each system contains blend-specific cross reactions that likely possess varied influences on the predicted flame speed.

Figure 2.45 presents the predicted flame speeds from the PMS models for the two valid blend cases, R290/R32 and R50/R32, though these PMS models have not yet reached the full 10-day run time on their reassigned, healthy compute nodes. Even at this early stage—having run for only 3 days—the PMS models should already provide a meaningful basis for comparison. Because the PMS core is seeded from the outset with all species present in the two converged pure-component mechanisms, the model begins its search already encompassing every reaction found in the corresponding concatenated mechanism, along with any cross-reactions that RMG's algorithm identifies between species originating from the two pure submechanisms. As such, even a partially completed PMS run should, in principle, capture the full chemistry of the concatenated approach while additionally incorporating some blend-specific interaction chemistry that concatenation alone cannot discover. Thus far, the PMS blends for both the R290/R32 and R50/R32 systems predict flame speeds that closely resemble the corresponding concatenated and MS blend models. The PMS models, which have run for approximately 3 days, remain immature but are still in progress.

In theory, the PMS blending method offers the most favorable balance of the three blending approaches. Unlike the concatenated method, PMS retains the ability to discover genuine blend-specific cross-reactions, since its core is built through a single, unified RMG exploration rather than the simple union of two independently generated mechanisms. At the same time, PMS avoids the slow, ground-up exploration required by the MS method: by seeding the initial core with species already known from the two converged pure-component mechanisms, PMS effectively begins its search much further along than MS does, allowing it to reach a comparably comprehensive core

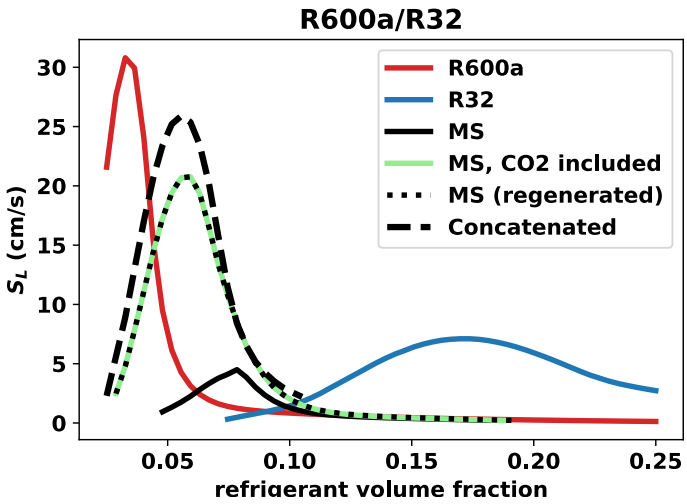


Figure 2.44: Predicted laminar burning velocities of pure R600a, pure R32, and 50/50 R600a/R32 mixtures by volume fraction combusted in normal air, with the x-axis representing total refrigerant volume fraction. Predicted flame speeds for the pure halocarbon mechanisms (solid red/blue), the concatenated blend mechanisms (dotted black), and the MS blend mechanisms at 97 core species lacking CO_2(solid black) and 98 core species including CO_2 (solid green), and the regenerated MS blend mechanism are plotted. Both MS blend models incorporated 97 species by the end of the 10-day run time.

within the same fixed run time budget. In this sense, PMS combines the chemical completeness advantage of concatenation with the cross-reaction-discovery capability of MS, without inheriting the primary drawback of either: it need not sacrifice run time to rediscover already-known pure-component chemistry, nor does it forgo the opportunity to identify new interaction pathways between blend components.

Model Type	System	Reactions	Species
Pure	R134a	1542	88
Pure	R152a	473	49
Pure	R290	4061	141
Pure	R32	3068	149
Pure	R50	3034	144
Pure	R600a	4858	164
Concatenated	R290/R134a	5527	210
Concatenated	R290/R152a	4452	171
Concatenated	R290/R32	6918	258
Concatenated	R50/R134a	4500	213
Concatenated	R50/R152a	3425	174
Concatenated	R50/R32	5891	261
Concatenated	R600a/R134a	6335	234
Concatenated	R600a/R152a	5249	194
Concatenated	R600a/R32	7769	287
MS	R290/R134a	2110	119
MS	R290/R152a	3807	149
MS	R290/R32	4639	163
MS	R50/R134a	3972	149
MS	R50/R152a	1261	174
MS	R50/R32	3713	143
MS	R600a/R134a	1154	97
MS	R600a/R152a	1200	79
MS	R600a/R32	1164	97

Table 2.4: Number of reactions and species in pure-component, concatenated blend, and minimally-seeded (MS) blend mechanisms.

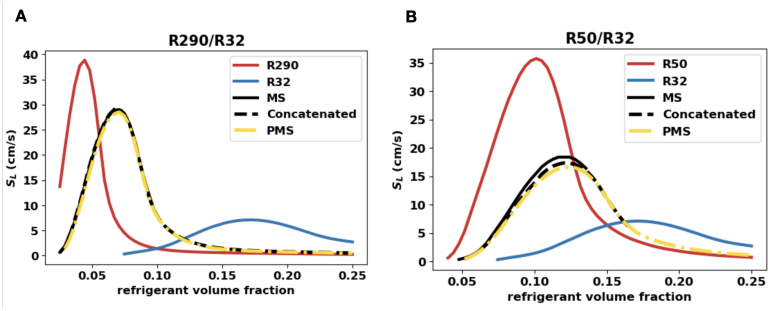


Figure 2.45: Predicted laminar burning velocities of **A)** pure R290, pure R32, and 50/50 R290/R32 mixtures by volume fraction combusted in normal air and **B** pure R50, pure R32, and 50/50 R50/R32 mixtures by volume fraction combusted in normal air, with the x-axis representing total refrigerant volume fraction. Predicted flame speeds for the pure halocarbon mechanisms (solid red/blue), the concatenated blend mechanisms (dotted black), and the MS blend mechanism (solid black), and the PMS blend mechanism (dash dot) are plotted. The PMS models, which have run for approximately 3 days, remain immature but are still in progress.

2.7 Correcting Node Regularization in RMG's Kinetic Family Decision Trees

2.7.1 Introduction

As described in Section 1.1.5, RMG constructs kinetic family decision trees using a machine-learning-based algorithm that iteratively splits nodes to cluster training reactions with similar rate coefficients [10]. During this automatic generation of family trees, a related step called *node regularization* is performed. In the context of decision trees related to machine learning, regularization refers to constraints that prevent trees from overfitting to the training data. Node regularization is intended to constrain each node's group definition so that it does not specify atom types more broadly than its training data actually supports—ensuring, for example, that a node is not permitted to match fluorine-containing reactions if no fluorinated reaction was ever included among its training set. Figure 2.46 provides a visual representation of node regularization in a decision tree, demonstrating how group definitions at specific nodes can change depending on whether regularization is performed or skipped. The regularization process essentially affects how far down training reactions can descend the tree. When using kinetic trees to estimate rate rules, the